

- (4) Increased by either, with joint effects being additive or multiplicative;
- (5) Reduced by the genotype and not affected by the environmental factor alone, but increased in the presence of both; or
- (6) Reduced by the genotype, increased by the environmental factor, and increased by the presence of both.

These models are simple and consider the effect of only one gene and one environmental factor. Interactions are likely to be much more complex, involving multiple genes, multiple environmental factors, genetic heterogeneity, and heterogeneity of exposure. However, finding statistical interaction between two factors is just the first step in unraveling a complex phenotype. Once statistical interactions are identified, other laboratory studies may be performed to establish what biological interactions, if any, exist.

The level of cholesterol in the bloodstream is an example of a trait that is caused by a complex set of genetic and environmental factors. In families with familial hypercholesterolemia (FHC), elevated cholesterol levels are inherited in a Mendelian, autosomal dominant pattern. However, only about 4 percent of all individuals in the top 5 percent of cholesterol levels in the population carry the gene responsible for FHC. Other genetic and environmental risk factors clearly influence cholesterol levels. For example, in people with and without FHC, consumption of cholesterol in the diet independently modifies cholesterol levels. Other factors such as exercise and medication use likely interact with dietary intake to determine blood levels of cholesterol. Other genetic factors may also be involved.

With the exception of a relatively small number of phenotypes that are completely genetically determined, almost all human phenotypes represent a combination of environmental and genetic factors. Understanding the ways in which genes and environment work together to impact human health is one of the great challenges in the study of complex phenotypes. SEE ALSO ALZHEIMER'S DISEASE; CARDIOVASCULAR DISEASE; COMPLEX TRAITS; DIABETES; INHERITANCE PATTERNS; METABOLIC DISEASE; STATISTICS.

William K. Scott

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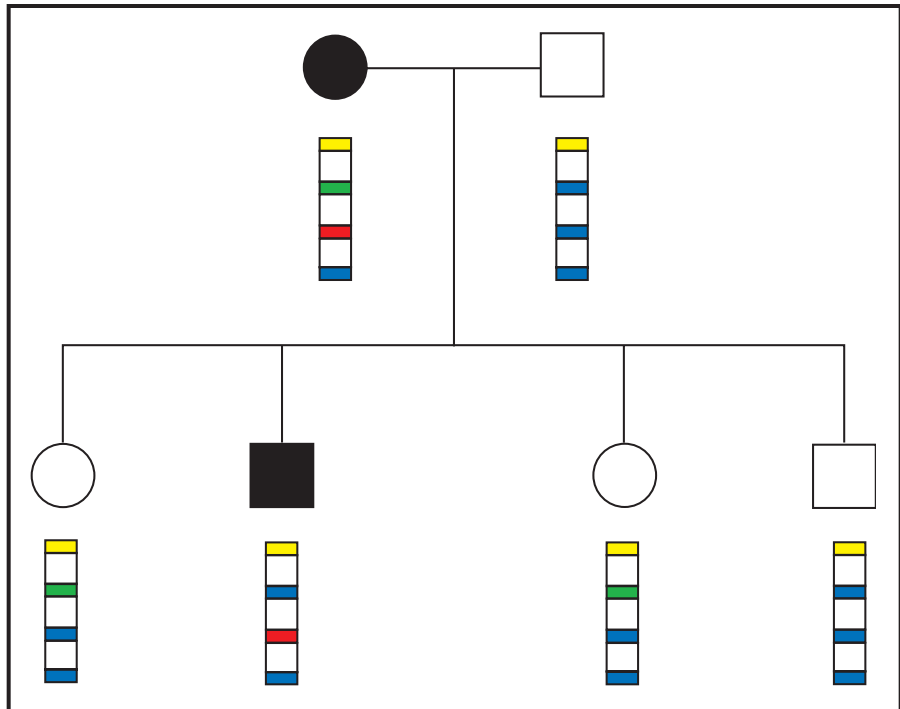
Gene Discovery

Gene discovery is the process of identifying genes that contribute to the development of a trait or **phenotype**. Researchers often try to discover the genes that are involved in specific diseases. They also try to find the genes that contribute to many other traits.

phenotype observable characteristics of an organism



Polymorphic markers along the chromosomes (here shown as different colored bars) are examined to determine which is coinherited with the disease (shaded circle and square). The red marker in the third position is found only in the two family members with the disease.



Gene discovery begins with clearly defining a trait of interest and determining if that trait has a genetic and/or environmental basis. This is done using several approaches, such as sibling recurrence risk ratio, familial aggregation, and twin and adoption studies. The sibling recurrence risk ratio is the frequency of a disease among the relatives of an affected person, divided by the frequency of the disease in the general population. The greater the ratio, the stronger the genetic component of the disease.

A trait also is suspected of having a strong genetic component when familial aggregation, which is the clustering of patients in a single family, occurs. Familial aggregation can sometimes be misleading, however. Since families often share the same environment, it is difficult to know whether environmental or genetic factors are the cause of clustering.

In twin studies, concordance rates play a critical role. Concordance is the percentage of second twins that exhibit a trait when the trait occurs in the first twin. Twins generally also share environments, so concordance rates are often compared between monozygotic twins, who are genetically identical, and dizygotic or “fraternal” twins, who share on average 50 percent of their genetic material. The greater the difference in concordance rates between monozygotic and fraternal twins, the stronger the genetic contribution to the trait. The combination of evidence from all these approaches indicates whether a particular phenotype is likely to have a genetic basis.

Approaches for Identifying Genes

Once genetic influence has been established, two research approaches are commonly used to identify the specific genes involved. These are the candidate gene approach and the genomic screening approach.

Candidate Gene Approach. In the candidate gene approach, genes are selected based on their known or predicted biological function and on their hypothesized relation to the disease or trait. These genes are subject to mutation analysis to determine whether they are really involved. The problem with the candidate gene approach is that it relies on assumptions about the molecular mechanisms underlying the development of a trait. However, diseases are usually studied because little is known about their causes, so initial ideas about these “molecular mechanisms” often prove to be wrong.

In addition, the candidate gene approach can be very time consuming, and it has been successful only infrequently. Chromosomal abnormalities, such as deletions, inversions, or **translocations**, in individuals exhibiting a trait, as well as animal models mimicking the trait, are especially important for a candidate gene approach, since they provide clues to the genetic basis of the trait.

Genomic Screen Approach. A genomic screen is a systematic survey in which **polymorphic** DNA markers, evenly spaced along all the chromosomes, are used to determine if a marker is inherited along with the trait, indicating genetic linkage. This is performed taking the DNA from each individual in the study and identifying the type of marker each has on his chromosomes. These data are then analyzed using statistical programs to see if the marker and the trait that is being studied travel together through families significantly more often than would be expected just by chance. If a DNA marker is found to be linked with a trait, it suggests that the marker and the gene responsible for the trait are rarely separated by crossing over and are therefore near each other on a region of a chromosome. Further fine mapping of this region with more closely spaced markers can narrow the region where the gene of interest lies.

Genomic screening does not require prior biological understanding of the pathophysiology of a disease. It requires large sets of data from families containing multiple members who are affected with the trait, and it tends to be the more expensive of the two approaches.

Genomic screening usually leads to the identification of one or several **loci**, or relatively large areas in the **genome**, that are linked with a trait but that contain many different genes. The genes in such regions need to be prioritized.

Genes are considered to be good candidates when their putative functions fit with the known or predicted pathway of the disease. If any known gene in the linkage region appears to be a good candidate gene, it is subjected to mutation analysis to determine if there is a potentially disease-causing mutation that segregates only with the affected individuals.

Both the gene candidate and the genetic screen approach require collecting data from a large number of families. Recruiting and medically evaluating affected and unaffected individuals for participation in a genetic study, and collecting their DNA samples, is a long and complicated phase of gene discovery.

Positional Cloning

Once one or more loci have been identified through a genomic screen as possibly containing a gene of interest, additional techniques are needed to

translocations movements of chromosome segments from one chromosome to another

polymorphic occurring in several forms

loci sites on a chromosome (singular, locus)

genome the total genetic material in a cell or organism



locate the exact gene responsible. Positional cloning is the process of identifying a disease gene based on its location on a chromosome. Genetic screening is the initial step of positional cloning. The usual steps are: (1) linkage (locates a chromosome area); (2) fine mapping (narrows down the initial genomic area to a smaller region); (3) candidate gene analysis (looks for mutations in genes lying in that small area). By using positional cloning, researchers can identify or “clone” a gene knowing only its location on a chromosome, as determined through linkage analysis. Once the location is identified, a physical representation of the genes and DNA in the linked region is constructed.

Before the Human Genome Project was completed, such a physical representation was constructed by using a “contig,” a group of overlapping DNA fragments that together cover the linked region. The contig is a scaffold or platform on which to place genes and other sequences in the correct position. The geneticist continues to collect families looking for new recombinations that reduce the piece of DNA that all the affected family members share but that has not been inherited by any of the unaffected individuals.

Positional cloning is a laborious process if the region is large and if the genes and **polymorphisms** making up the contig are not known. This portion of the process has been greatly helped by the Human Genome Project, as it provides the contig all filled out and correctly mapped.

In most cases, the smallest inherited piece of DNA is still quite large molecularly, and can contain many genes. Only one gene causes the disease, though, so each gene must be tested for mutations that segregate with the trait. This can be very time consuming as well. If no mutations are found, the process is repeated with the next gene in the smallest shared region of DNA.

The third gene identified through positional cloning was the cystic fibrosis transmembrane conductance regulator gene. The gene, identified in 1989, regulates chloride ion transport across the plasma membrane and consists of twenty-seven **exons** spread over a 230-kilobase-pair region on chromosome seven. This gene was difficult to clone because of the lack of chromosome abnormalities that would have helped locate it. No human genome sequence was available at that time. Numerous technical problems also arose in constructing a physical map of the linkage region, creating the need to screen numerous libraries to obtain a full-length clone of the gene. More than 550 mutations were identified by late 2001.

Complex Diseases

The sequencing of the human genome and the advancement of **bioinformatics** and molecular tools to identify genes has made the process of gene discovery much easier. Most of the human genome sequence is now available on the Internet, with known and predicted genes annotated. This has reduced the need to laboriously build contigs.

Many of the common single-gene diseases have already been associated with a gene, and research efforts are now shifting to the more difficult task of finding genes that give susceptibility to developing complex diseases. (Such diseases show familial aggregation but do not follow any clear Mendelian inheritance pattern.)

polymorphisms occurring in several forms

exons coding regions of genes

bioinformatics use of information technology to analyze biological data

It is thought that a combination of several genetic predisposition factors interact with environmental factors to trigger complex diseases, and that it is not a single gene but multiple genes that contribute to such traits, and the identification of each of these genes is correspondingly more difficult.

Complex traits can constitute various other challenges for researchers. Genetic heterogeneity is where **alleles** at more than one locus trigger the same phenotype, or mutations in the same gene cause different phenotypes. Reduced penetrance is where a predisposing genotype does not necessarily cause the phenotype to manifest itself. Phenocopy is where a trait looks identical but has a different cause than the one being studied.

To address these challenges, scientists use association studies, which are based on the principle that if a particular allele and trait occur simultaneously at a statistically significant frequency, the allele is likely to be involved in the development of the trait. (Linkage studies, by contrast, are based on finding DNA markers and traits that are linked within families.)

Alzheimer's disease is one example of a complex trait. Three genes have been found to contribute to the rare, early forms of the disease. Genetic screens have found that a fourth locus is linked to the common, late-onset form of the disease. Association studies have revealed that one allele of this fourth locus increases a patient's risk of developing Alzheimer's in a dose-dependent fashion, where the risk posed by having two alleles is greater than the risk posed by having just one. SEE ALSO ALZHEIMER'S DISEASE; BIOINFORMATICS; CLONING GENES; COMPLEX TRAITS; CYSTIC FIBROSIS; DNA LIBRARIES; HUMAN GENOME PROJECT; INTERNET; LINKAGE AND RECOMBINATION; MENDELIAN GENETICS; TWINS.

Sofia A. Oliveria and Jeffery M. Vance

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Gene Expression: Overview of Control

The chromosomes of an organism contain genes that encode all of the RNA and protein molecules required to construct that organism. Gene expression is the process through which information in a gene is used to produce the final gene product: an RNA molecule or a protein.

Each cell in a multicellular organism such as a human contains the same genes as every other cell. Nonetheless, there are hundreds of distinct types of cells in the human body, each expressing a unique set of genes. Indeed, it is this unique constellation of expressed genes that makes each cell type distinct.

Cells may also change the genes they express over time, and they are constantly adjusting the amount of protein made in response to changing conditions. How does a cell express some, but not all, of the genes in its **genome**? How does it react to environmental changes to adjust the level of

alleles particular forms of genes

genome the total genetic material in a cell or organism



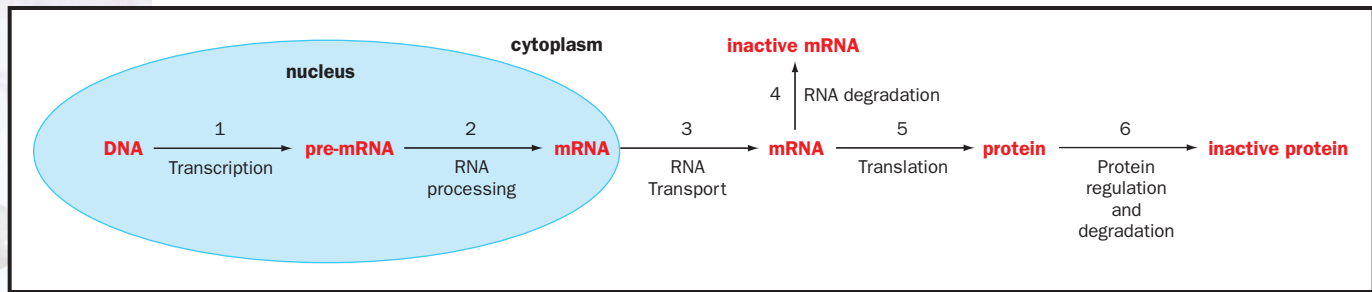


Figure 1. The flow of genetic information from DNA to proteins. Control can be exerted at each numbered step. Most control occurs at step one, transcription.

transcription messenger RNA formation from a DNA sequence

cytoplasm the material in a cell, excluding the nucleus

phosphorylation addition of the phosphate group PO_4^{3-}

gene expression? These are the problems of control of gene expression. While the genes whose final products are RNA molecules are also regulated, this entry will focus on genes that encode proteins.

The Flow of Genetic Information from Genes to Proteins

Cells can regulate gene expression at every step along the way, from DNA to the final protein, as shown in Figure 1. Genetic information in DNA is first copied to form an RNA molecule, in a process known as **transcription**. The RNA used to make proteins is called messenger RNA (mRNA) because it carries information from the DNA to the ribosome, where protein synthesis occurs.

The mRNA serves as a template to guide protein synthesis. Scientists refer to protein synthesis as “translation” because ribosomes translate an mRNA sequence into a protein sequence. Prokaryotic cells use the mRNA directly as a template for protein synthesis.

Eukaryotic cells, however, must modify the precursor mRNA in several ways before it can be used to guide protein synthesis. The two ends are chemically altered, and sections of the RNA that do not encode protein sequences, called introns, are spliced out. Together, these modifications are called “mRNA processing.” After processing, the mature mRNA moves from the nucleus into the **cytoplasm**, where it binds to the ribosome and serves as a template for synthesis of a protein.

The most important stage for the regulation of most genes is when transcription begins. This is because it costs the cell less energy to regulate transcription than to regulate the steps after transcription. The second point where regulation occurs is during RNA processing. Cells can regulate the rate of processing. In addition, the final mRNA product can be altered through alternative splicing, as shown in Figure 2. Alternative splicing can regulate the types of proteins produced from a single gene.

Cells can also regulate mRNA transport out of the nucleus. Once the mRNA has moved into the cytoplasm, the abundance of mRNA can be regulated by RNA degradation. Cells can regulate translation, controlling the number of proteins each mRNA produces. Finally, even after a cell has generated a protein, it can regulate the abundance and activity of that protein. For instance, cells regulate the activity of many proteins by post-translational modifications such as **phosphorylation**. Cells can also regulate the abundance of most proteins by degrading them.

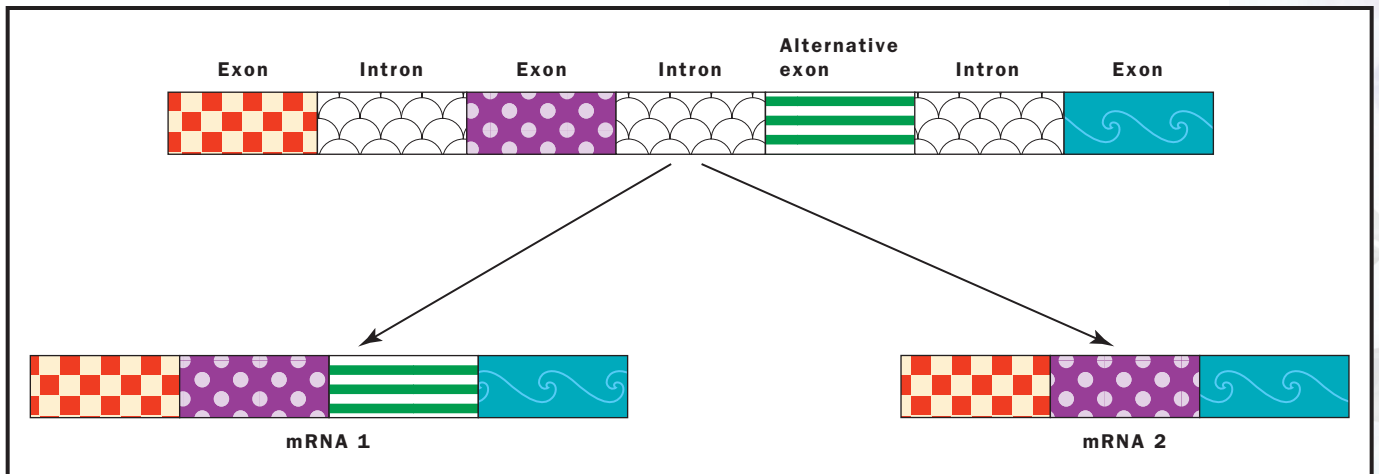


Figure 2. Alternative mRNA splicing leads to isoforms, or related proteins.

RNA polymerase
enzyme complex that creates RNA from DNA template

chromatin complex of DNA, histones, and other proteins, making up chromosomes

transcription factors
proteins that increase the rate of gene transcription

Gene Control Occurs at Several Levels

For a gene to be transcribed, **RNA polymerase** must first find the gene. This is made more difficult by the tight packing required to fit the entire genome within the nucleus. The cell uses this packing to its advantage, though, to prevent access to and expression of genes in some chromosomal regions.

Regions that are tightly condensed are called heterochromatin and can be distinguished from more open regions (euchromatin) by their dense staining, when viewed under a microscope. In females, an entire X chromosome in each cell is kept condensed throughout life, to avoid a “double dose” of these genes (recall that females have two X chromosomes, while males have only one). This random X inactivation leads to mosaicism, in which some female cells express genes from one chromosome, while others express genes from the other.

Even within an active chromosome, some regions may be temporarily inactivated. Inactive heterochromatin can be converted to active euchromatin, and vice versa, by chemical modification of the histone proteins to which the DNA is attached in the chromosome. Negatively charged DNA is chemically attracted to the positively charged histones. By adding or removing chemical groups to the histones, this attraction can be modulated. A weaker attraction, as would occur by adding negatively charged groups to the histones, tends to open up the **chromatin**, favoring gene expression. A stronger attraction keeps it more condensed.

How Do Cells Regulate Transcription?

To understand transcriptional regulation, consider the structure of a typical eukaryotic gene, shown in Figure 3. The promoter of a gene is the binding site for a group of general **transcription factors** and for RNA polymerase. Transcription begins when a complex of proteins called TFIID binds to a promoter. The sequential binding of other general transcription factors and RNA polymerase follows. A protein tail tethers the RNA polymerase to the general transcription complex. When the general transcription factor TFIIH phosphorylates this tail, the RNA polymerase is released and moves along the DNA to begin transcription.

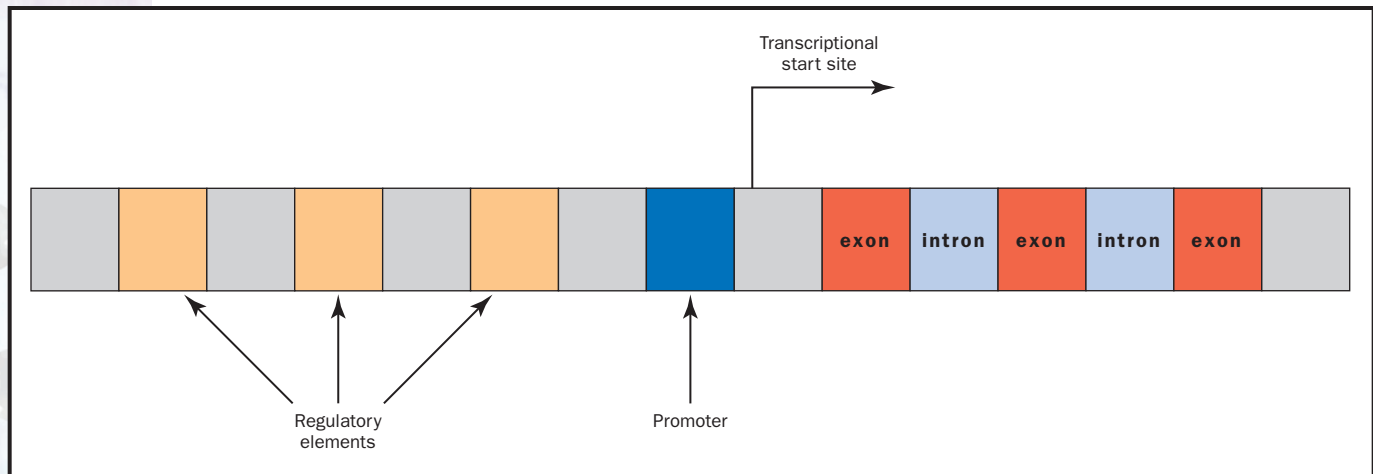


Figure 3. Structure of a typical eukaryotic gene.

These steps are identical for essentially all genes. However, the rate at which each of the steps occurs can be influenced by the presence or absence of other, gene-specific transcription factors. It is these other factors, often called gene regulatory proteins, that give the cell the ability to turn some genes on and others off. In contrast to the few general transcription factors, which assemble on the promoters of all genes, there are thousands of gene regulatory proteins. The regulatory proteins bound to a gene vary from gene to gene, and each is usually present at low levels in the cell.

Binding sites for these additional regulatory proteins are usually located upstream from the promoter. Surprisingly, these binding sites can be located some distance from the promoter and still regulate transcription. It is thought that this action at a distance can occur because the DNA between the regulatory sequence and the promoter can loop out, to allow the regulatory protein to contact the promoter, as shown in Figure 4.

Regulatory proteins are classified as either activators or repressors. Activators increase the rate of transcription, whereas repressors decrease it. The DNA sequences that bind activator proteins are called “enhancer elements,” and those that bind repressor proteins are “repressor elements” or “silencer elements.”

Many regulatory proteins have at least two distinct regions or domains, as shown in Figure 5. One domain binds to a specific DNA sequence. The other domain typically contacts the general transcription machinery assembled at the promoter. In one class of activating proteins, the activation domain contains a cluster of negatively charged (acidic) amino acids. Scientists believe acidic transcriptional activators accelerate the assembly of general transcription factors on the promoter. This is just one way a transcriptional activator can work. For instance, other regulatory proteins affect how tightly the gene is packaged within chromatin. Opening up the chromatin allows the transcription machinery to more quickly gain access to the promoter.

Gene regulatory proteins bind to the DNA when it is in a double-helical state. They recognize a specific DNA sequence by forming hydrogen bonds to chemical groups on the outside of the DNA. These proteins often contain common identifiable structural “motifs” that directly contact specific DNA sequences.

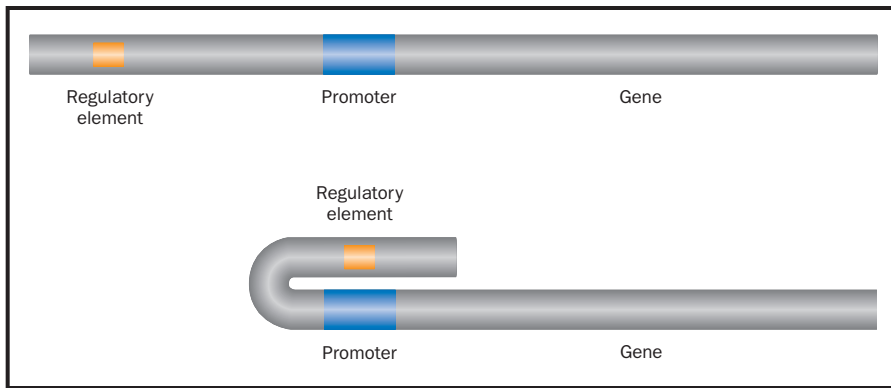


Figure 4. A regulatory element contacting the promoter by looping out the intervening DNA.

Combinatorial Regulation of Gene Expression

Some eukaryotic gene regulatory proteins work individually, but most act within a complex of proteins. Furthermore, a single gene regulatory protein may participate in multiple types of regulatory complexes. For example, a protein might function in a complex that activates the transcription of one gene and in a complex that represses transcription of another gene.

A gene that must be turned on at different times and in different tissues during development might have gene regulatory proteins clustered at multiple sites along its regulatory region. These complexes can then regulate the expression of the gene in a variety of developmental processes. The rate of RNA synthesis initiation will depend on the combination of regulatory proteins bound to the control regions of the gene. Thus, we can think of the regulatory region of the gene as an information processor, like a computer, that integrates input from all the regulatory proteins present and determines an appropriate level of RNA synthesis.

Regulation of Gene Expression during Development

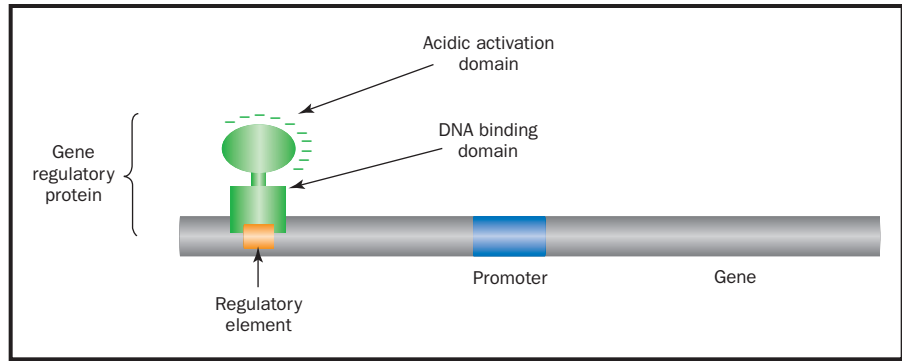
During development, cells become different from one another because they synthesize and accumulate different proteins. Most of these differences come from changes in gene expression. Specialized cell types result from different genes being turned on or off in a coordinated manner. When a cell becomes a specific cell type, it continues in this role through many subsequent cell generations. This implies that cells remember the changes in gene expression involved in the choice of cell type. How is this achieved? One way is through the euchromatin-heterochromatin conversion, which can be faithfully inherited through cell division. Another way is for an important regulatory protein to activate its own expression as well as the expression of other genes. This gene, once expressed, will maintain its own expression.

As described above, it is usually a combination of gene regulatory proteins, rather than a single protein, that determines when and where gene expression occurs. Certain proteins can be more important than others, though. If all the other factors required for expression of a group of genes are present, a single gene regulatory protein can switch a cell from one developmental pathway to another.

For example, forced expression of the MyoD protein in **fibroblasts** will cause these cells to form into muscle fibers. In an extreme example, homeotic

fibroblasts undifferentiated cells that normally give rise to connective tissue cells

Figure 5. Domain structure of an acidic activation protein. This protein enhances the ability of general transcription factors to bind to the promoter.



genes specify large regions of an animal's body plan. Mutations in these genes can transform one body part into another. For instance, a mutation in the *antennapedia* gene of fruit flies will convert antennae to legs. Thus, expression of a single gene can trigger the expression of a whole battery of genes.

Combinatorial Control

An advantage of multiple gene regulatory proteins over single ones is that many different genes can be controlled with a handful of proteins. Consider the opposite situation, in which every gene would need a unique regulatory protein. The gene for each of those proteins would also need its own protein, and so on. Instead, a smaller set of proteins combines in different ways to make a large set of regulatory possibilities. Imagine that any regulatory element must be composed of two proteins. Four proteins (a, b, c, and d) can combine in pairs in ten different ways (aa, ab, ac, ad, bb, bc, etc.), and could thus, theoretically, control ten different genes. In reality, the situation is more complex, with literally thousands of regulatory proteins combining in ways researchers have not even begun to calculate.

With combinatorial control, therefore, a regulatory protein does not necessarily regulate a particular battery of genes or specify a particular cell type. Instead it might serve many purposes, and those purposes might overlap with those of other regulatory proteins. A regulatory protein might be switched on in many cell types, at different locations in the animal, and several times during development. Thus, combinatorial gene control makes it possible to generate a great deal of biological complexity with relatively few gene regulatory proteins.

Hormones and Growth Factors

During development, cells can change the expression of their genes when influenced by both external and internal signals. Signals from outside the cell that influence gene expression include contact with other cells, growth factors, and **hormones**.

Growth factors are extracellular molecules that stimulate a cell to grow or proliferate. Examples include epidermal growth factor and fibroblast growth factor. Growth factors regulate gene expression indirectly through a network of intracellular signaling cascades.

Hormones are signaling molecules that endocrine cells secrete into the bloodstream. Some hormones, such as insulin, bind to cell surface receptors

hormones molecules released by one cell to influence another

and affect gene expression through a network of intracellular signaling cascades. Other hormones, such as testosterone, pass through the cell membrane and bind to regulatory proteins in the cell that directly regulate transcription.

When the Regulation of Gene Expression Fails

When the control of gene expression fails, there can be serious consequences, such as death, birth defects, and cancer. Birth defects can result when the regulation of one or more genes important for development is lost. This often occurs because of a mutation, but it can also occur if the embryo or fetus is exposed to certain chemicals, such as alcohol. Mutations in the receptor for fibroblast growth factor, for instance, cause dwarfism. Cancer occurs when the regulation of genes that control growth and cell division, programmed cell death (apoptosis), and cell migration are lost. SEE ALSO ALTERNATIVE SPLICING; BIRTH DEFECTS; DEVELOPMENT, GENETIC CONTROL OF; GENE; HORMONAL REGULATION; POST-TRANSLATIONAL CONTROL; PROTEINS; RNA PROCESSING; SIGNAL TRANSDUCTION; TRANSCRIPTION; TRANSCRIPTION FACTORS.

Eric Aamodt

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Gene Families

Gene families are groups of DNA segments that have evolved by common descent through duplication and divergence. They are multiple DNA segments that have evolved from one common ancestral DNA segment that has been copied and changed over millions of years.

The members of a gene family may include expressed genes as well as nonexpressed sequences. Such nonexpressed sequences include promoters, operators, transposable genetic elements, and pseudogenes, which are genes that are no longer functionally expressed.

Pseudogenes resemble other family members in their linear sequence of nucleotides. However, they usually either lack the signals that would allow them to be expressed or have significant deletions or rearrangements that prevent successful transcription or translation.

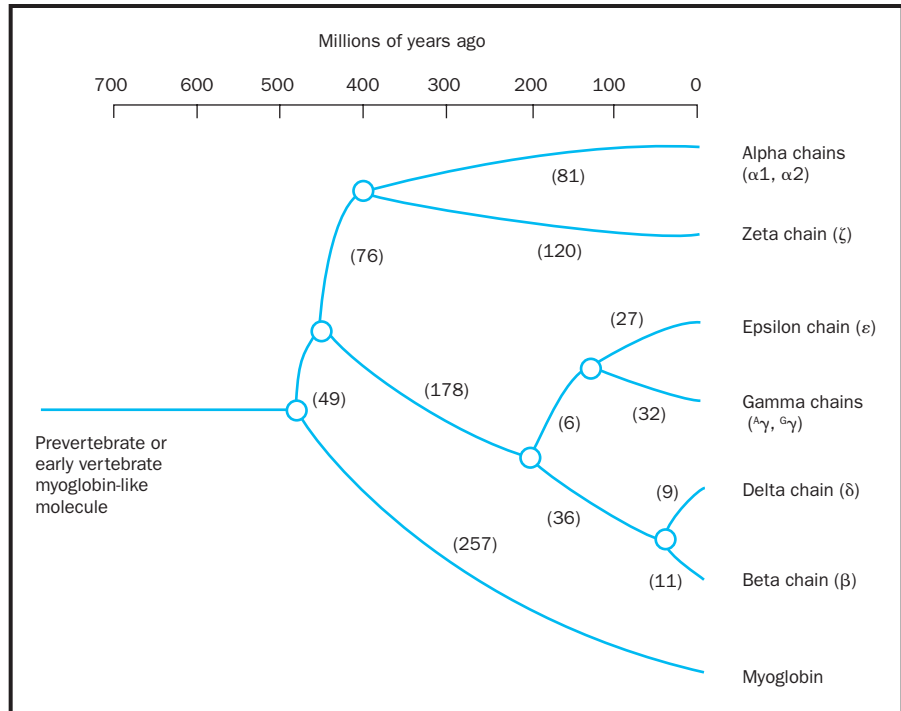
One well-studied gene family is that of the globins, shown in both Figures 1 and 2. The globin family contains many pseudogenes as well as many functional genes, including the genes coding for hemoglobins (α , β , γ , δ).

Gene families vary enormously in size and number, ranging, in the human **genome**, from just a few copies of very closely related sequences to more than a half-million copies of Alu sequences, which are transposable

genome the total genetic material in a cell or organism



Figure 1. Evolution of the globin genes. The numbers in parentheses represent the estimated number of nucleotide changes needed to account for the observed amino acid differences. Adapted from <<http://www.cord.edu/faculty/landa/courses/b315f99/sessions/phylogeny/globinPhylogeny.jpg>>.



genetic elements with no known function. For genes that encode proteins, duplicate copies of genes have been found for over 2,000 proteins in a variety of genomes.

Members of gene families may be located in contiguous clusters on one chromosome, or they may be scattered throughout a genome. Homeotic genes, which lie in contiguous clusters on a few chromosomes, provide the best example of evolutionarily preserved gene order within a gene family. These genes play a role in the spatial development of the anterior-posterior axis of vertebrates and invertebrates. Four binary axes are laid down early in the **basal** body plan of most metazoans, including us: anterior-posterior, dorsal-ventral, left-right, and inside-outside. They affect locations on the body axis in roughly the same order as they are arranged along a chromosome, even though different clusters appear on different chromosomes. Amazingly, these genes work about the same in a fruit fly as they do in a human in establishing linear arrangements.

In some gene families, related genes have stayed together over long periods of evolution, while in other gene families, members have become widely distributed within genomes. In ribosomal RNA genes, tandem arrays, in which multiple copies of the same gene occur one after another, have been observed. On the other hand, in the globin gene family, both the order and distribution of the genes, which are not identical, vary widely, even within one taxonomic group such as the mammals.

Members of a gene family usually have similar structures, but they may have diverged evolutionarily to such an extent that they are expressed in different ways. They may be expressed at different times in the development of multicellular organs or in different cells and tissues, and they may have acquired different functions. They may even have been transferred between organisms that are not closely related by evolution.

basal lowest level

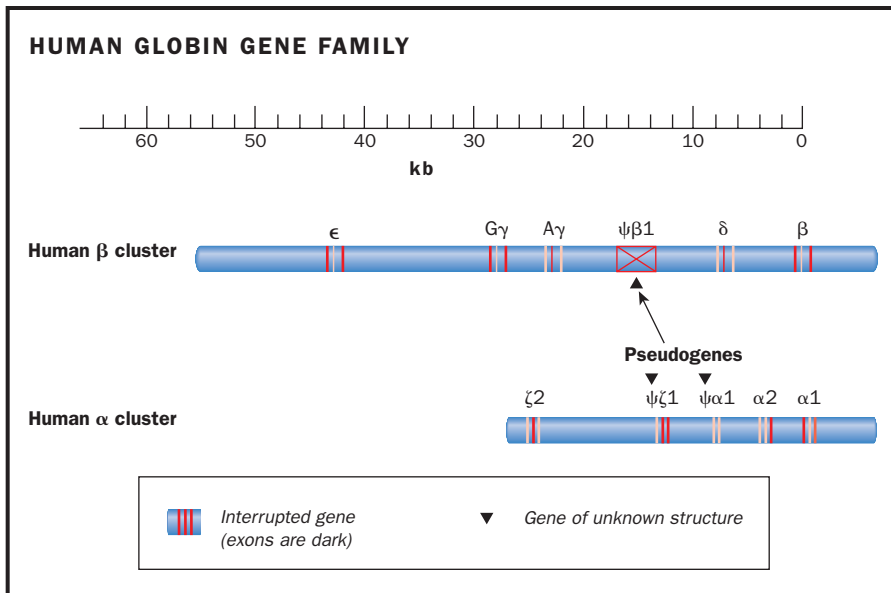


Figure 2. Representations of the two clusters of genes that code for human globin, the protein portion of hemoglobin. Adapted from <<http://www.irm.pdx.edu/~newmanl/GlobinGeneEvolution.GIF>>.

One controversy about gene families involves whether they have arisen primarily by **polyploidy** or via tandem gene duplications. Polyploidy means that full genomes in an organism are duplicated either by mitosis or meiosis without cytokinesis or by matings between organisms with unequal numbers of chromosomes. This is followed by full copying of both parents' full genomes so each haploid set of chromosomes is now diploid. In tandem duplication, one or more copies of a gene lie on the same chromosome adjacent to one another. Polyploidy has been invoked to explain the evolution of complex new functions in taxa. Researchers give five reasons. Polyploids:

- (1) have "higher levels of heterozygosity than do their diploid parents";
- (2) "exhibit less inbreeding depression than do their diploid parents";
- (3) "are polyphyletic . . . [which] . . . incorporates genetic diversity from multiple progenitor populations" and, thus, they have higher genetic diversity "than expected by models of polyploid formation involving a single origin";
- (4) have genome rearrangements that are common; and
- (5) are like duplicated genes, freed from intense selection pressure, which allows frequent evolution of new functions (Soltis and Soltis 2000, p. 310).

Austin Hughes has been the major critic of the often invoked polyploid hypothesis for origin of major animal groups because of the major substitutional load that would be involved and molecular phylogenetic evidence against it (1999, pp. 205–212). However, the results of most molecular evolutionary studies are more consistent with the gradualist view that new functions are generated primarily by tandem gene duplication and divergence of both sequence and function, spread over a long time. SEE ALSO DEVELOPMENT, GENETIC CONTROL OF; EVOLUTION OF GENES; POLYPLOIDY; PSEUDOGENES; TRANSPOSABLE GENETIC ELEMENTS.

John R. Jungck

polyploidy presence of multiple copies of the normal chromosome set

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Internet Resources

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Gene Flow

Gene flow is the transfer of genetic material between separate populations. Many organisms are divided into separate populations that have restricted contact with each other, possibly leading to reproductive isolation. Many things can fragment a species into a collection of isolated populations. For example, a treacherous mountain pass may cut off one herd of mountain goats from another. In human beings, cultural differences as well as geographic separation maintain unique populations: It is more likely that a person will marry and have children with someone who lives nearby and speaks the same language.

Over time, reproductive isolation can lead to genetic differences between two populations. Gene flow between populations limits this genetic divergence, serving to inhibit the development of separate species out of the two separated populations.

The essential mechanism of gene flow is movement of individuals (or their **gametes**) between populations. For example, gene flow can occur in plant species when pollen is carried by bees or blown by the wind from one population of flowering plants to another.

Migration has been a significant feature of human history in both prehistoric and more recent times. No gene flow occurs if an individual migrates into a different population but does not reproduce. The migrant's genes must become part of the genetic makeup of the population into which it has migrated.

In most populations, not all individuals contribute equally to the next generation. Because each individual can have different alleles, when only a subset of individuals reproduce, allele frequencies change from generation to generation, and some alleles may be lost. A change in allele frequency due to random chance is known as **genetic drift**, whereas a change due to differ-

WHAT IS AN ALLELE?

Alleles are forms of a gene that may differ between individuals or populations; brown and blue eye colors are due to different alleles for eye color.

gametes reproductive cells, such as sperm or eggs

genetic drift evolutionary mechanism, involving random change in gene frequencies

ences in reproductive fitness is known as natural selection. Gene flow between isolated populations slows down their genetic drift from each other and reduces the power of natural selection to promote divergence between them. When there is a great deal of gene flow between populations, they tend to be similar; in this way, gene flow has a homogenizing effect. The opposite also tends to be true: If there is little or no gene flow between populations, the genetic characteristics of each population are more likely to be different.

Gene flow does not just occur between two populations. When a series of populations exists over a large area, gene flow may serve to keep even the most distant populations similar to one another. This can occur even if they do not exchange individuals or gametes as long as the alleles from one population eventually flow into the other population through a series of migrations or gamete movements. Similarly, other types of separation can also be overcome by this type of graded gene exchange. For instance, Great Danes and Chihuahuas cannot breed directly because of size incompatibility. But gene flow in both directions, through intermediate-sized dogs, keeps these two breeds from becoming separate species.

It is very difficult to assess gene flow directly, so population geneticists have devised a way to estimate gene flow by comparing allele frequencies. By determining allele frequencies in two different populations, the amount of gene flow between them, usually expressed as the number of migrants exchanged per generation, can be estimated. SEE ALSO GENETIC DRIFT; HARDY-WEINBERG EQUILIBRIUM; POPULATION GENETICS.

R. John Nelson

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Gene Targeting

Gene targeting is a method for modifying the structure of a specific gene without removing it from its natural environment in the chromosome in a living cell. This process involves the construction of a piece of DNA, known as a gene targeting **vector**, which is then introduced into the cell where it replaces or modifies the normal chromosomal gene through the process of **homologous** recombination.

vector carrier

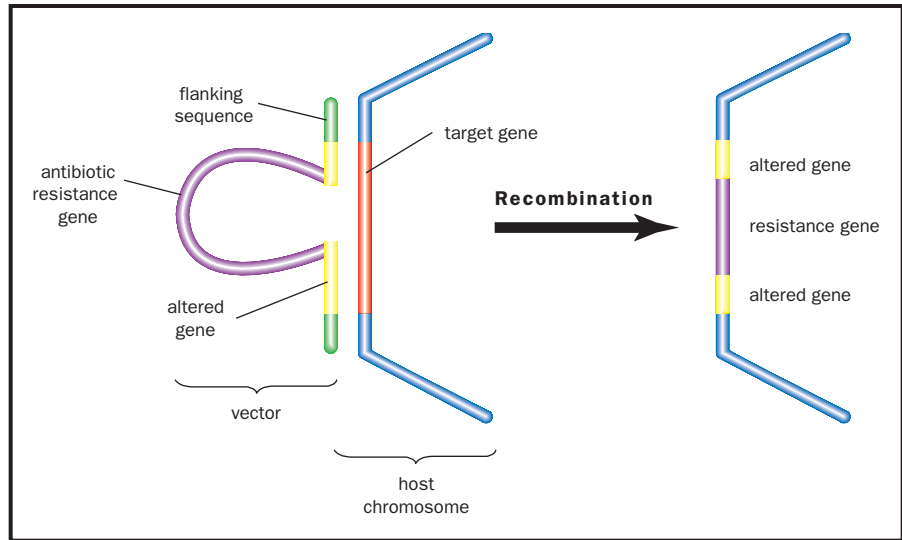
homologous carrying similar genes

The Homologous Recombination Process

Homologous recombination is a process that occurs within the chromosome and which allows one piece of DNA to be exchanged for another piece. It is a cellular mechanism that is probably part of the normal process cells use to repair breaks in their chromosomes. Homologous recombina-



A modified version of the target gene replaces it in the chromosome. The target gene is removed and degraded. In this example, the gene is modified by insertion of an antibiotic resistance gene, which both inactivates the gene and allows efficient selection of transformed cells.



tion requires that the pieces of DNA undergoing recombination be almost identical (homologous) in sequence. In addition, sequences on either side of the target should be identical, to promote more efficient targeting and recombination.

By constructing a sequence that is homologous to a target sequence (such as a gene), laboratory researchers can replace one of the cell's own copies of a particular gene with a copy that has been altered in some way. It is also possible to replace only a part of a gene, such as one portion of its protein coding region. This permits the introduction of a **mutation** into specific cellular genes, which can either stop the gene functioning altogether (called a “**knock out**”) or can mimic changes to genes that have been implicated in human diseases. The ability to target DNA constructs to particular locations in chromosomes is a very powerful tool because it allows the modification of more or less any gene of interest, in more or less any way desired.

Homologous recombination of a DNA vector into a gene of interest can be done in almost any cell type but occurs at a very low frequency, and it is therefore important to detect the few cells that have taken up the gene. Gene targeting vectors are designed with this in mind. The simplest strategy is to include an antibiotic resistance gene on the vector, which interrupts the sequence homologous to the gene of interest and thus makes the inserted gene nonfunctional. This “selectable marker” gene makes the cells that possess it resistant to antibiotics, and can then be used to eliminate cells that are not genetically modified.

An example of a selectable marker that is commonly used for this purpose is the puromycin-N-acetyl-transferase (*pac*) gene, which confers resistance to the antibiotic puromycin, a drug that inhibits the function of **ribosomes**. After the introduction of the DNA construct, the cells are cultured with puromycin in the **medium**. This allows the selection of single cells that have incorporated the DNA construct into their own chromosomes. Cells lacking the *pac* gene will die in a culture medium containing puromycin. Once the puromycin resistant cells have been expanded into cell lines, the DNA of these cells can then be analyzed to select out a subset of the cells in which the introduced construct has integrated into the correct (target) gene.

mutation change in DNA sequence

knock out deletion of a gene or obstruction of gene expression

ribosomes protein-RNA complexes at which protein synthesis occurs

medium nutrient source

For reasons that are not yet fully understood, the rate of homologous recombination in mouse embryonic stem (ES) cells is substantially higher than that of most other cells. Once a clone of ES cells with the correct targeting event has been identified, these cells can be used to introduced into the mouse via the process of **blastocyst** injection, which allows the study of gene function in the bodies of living, intact animals. Until very recently mice were the only organisms in which it has been possible to introduce targeted mutations into the germ line. The development of nuclear transfer (moving the nucleus from one cell to another), however, has allowed gene targeting to be done in other mammalian species, such as sheep and pigs.

blastocyst early stage of embryonic development

Adding or Deleting Genetic Material

As well as mutating or knocking out specific genes, gene targeting allows the introduction of novel pieces of DNA into a specific chromosomal location (this is often termed a “knock-in”). This allows researchers to examine the function of a gene in a variety of ways. For example, it is possible to examine where in the animal the gene is normally expressed by insertion (knock-in) of a fluorescent protein (such as green fluorescent protein, GFP) into the gene so that the cells expressing the gene begin to glow. In addition to changing single genes it is also possible to remove or alter large pieces of chromosomes.

Technologies also now exist that allow genes to be removed not just in a whole animal, as described above, but in a subset of cells or in a particular tissue. This can be achieved by modifying the vector to include target sites (termed loxP sites) for an enzyme called Cre recombinase. When the Cre enzyme is present in a mouse cell in which the target gene is surrounded by loxP sites, it will cut this gene out of the chromosome. This allows the function of this gene, which may be required for the mouse to normally develop, to be analyzed in a particular cell type or tissue where only the Cre recombinase is expressed.

Therapeutic Potential of Gene Targeting

It is hoped that gene targeting may eventually become useful in treating some human genetic disorders such as hemophilia and Duchenne muscular dystrophy. Treating human disease by the types of genetic approaches mentioned above is termed “gene therapy.” This could, in principle, be achieved by replacing the defective gene with a normal copy of the gene in the affected cells of an individual undergoing treatment. In order to make this potential treatment effective it will be necessary to develop technologies that increase the frequency with which targeting occurs. This is currently the subject of much research.

The development of nuclear transfer technology also has opened up the possible alternative method of using homologous recombination for gene therapy based on cell transfer. Gene targeting would be used to replace the defective genes in selected **somatic** cells in culture, and their nuclei could then be transferred into stem cells. The stem cells can then be differentiated into the affected cell type (for example, into bone marrow cells for hemophilia) and these cells could then be transplanted to patients. SEE ALSO EMBRYONIC STEM CELLS; GENE; GENE THERAPY; MARKER SYSTEMS.

somatic nonreproductive; not an egg or sperm

Seth G. N. Grant and Douglas J. C. Strathdee



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Gene Therapy

Gene therapy is a new and largely experimental branch of medicine that uses genetic material (DNA) to treat patients. Researchers hope one day to use this therapy to treat several different kinds of diseases. While rapid progress has been made in this field in recent years, very few patients have been successfully treated by gene therapy, and a great deal of additional research remains to be done to bring these techniques into common use.

Disease Targets

Humans possess two copies of most of their genes. In a recessive genetic disease, both copies of a given gene are defective. Many such illnesses are called loss-of-function genetic diseases, and they represent the most straightforward application of gene therapy: If a functional copy of the defective gene can be delivered to the correct tissue and if it makes (“expresses”) its normal protein there, the patient could be cured. Other patients suffer from dominant genetic diseases. In this case, the patient has one defective copy and one normal copy of a given gene. Some of these disorders are called gain-of-function diseases because the defective gene actively disrupts the normal functioning of their cells and tissues (some recessive diseases are also gain-of-function diseases). This defective copy would have to be removed or inactivated in order to cure these patients.

Gene therapy may also be effective in treating cancer or viral infections such as HIV-AIDS. It can even be used to modify the body’s responses to injury. These approaches could be used to reduce scarring after surgery or to reduce restenosis, which is the reclosure of coronary arteries after balloon angioplasty. Each of these cases will be discussed in more detail below, but first we will deal with two technical issues of gene transfer: gene delivery and longevity of gene expression.

Gene Delivery

Whether given as pills or injections, most conventional drugs simply need to reach a minimal level in the bloodstream in order to be effective. In gene therapy, the drug (DNA) must be delivered to the nucleus of a cell in order to function, and a huge number of individual cells must each receive the DNA in order for the treatment to be effective. The situation is further complicated by the fact that a given gene may normally function in only a small portion of the cells in the body, and **ectopic expression** may be toxic. Thus, successful gene therapy often requires highly efficient delivery of DNA to a very restricted population of cells within the body.

ectopic expression
expression of a gene in
the wrong cells or tis-
sues



To achieve these goals, many researchers have turned to viruses. Viruses are parasites that normally reproduce by infecting individual cells in the human body, delivering their DNA to the nucleus of those cells. Once there, the viral DNA takes over the cell, converting it to a factory to make more viruses. The cell eventually dies, releasing more viruses to continue the cycle. Scientists can remove or disable some of the genetic material of the virus, making it unable to reproduce outside of the laboratory. This genetic material can then be replaced by the gene needed to treat a patient. The modified (or recombinant) virus can then be administered to the patient, where it will carry the therapeutic gene into the target cells. In this way, scientists can take advantage of the virus's ability, gained over millions of years of evolution, to deliver DNA to cells with tremendous efficiency. One of the most commonly used is a cold virus called adenovirus. Recombinant adenoviruses have been used in experimental gene therapy for muscle diseases, and can deliver genes to almost all of the cells in a small region surrounding the site of injection. Unfortunately, while adenoviruses excel at gene delivery, evolution is a double-edged sword, and the many mechanisms our own bodies have evolved to combat harmful viral infections are also used against therapeutic viruses, as will be discussed in more detail below.

Recombinant adenoviruses cannot be used to transfer DNA to all cell types, because they cannot reproduce themselves outside of the laboratory.

In 1999 doctors at the Ohio State University Medical Center prepare a gene therapy injection for 36-year-old Donovan Decker. He is the first patient to ever receive gene therapy for muscular dystrophy.

In April 2002 researchers announced that *ex vivo* gene therapy for severe combined immunodeficiency had been successful in five boys for up to 2.5 years.

precursor a substance from which another is made

ex vivo outside a living organism

vectors carriers

When a cell with a recombinant adenovirus in it divides, only one of the two resulting cells contains the virus and the therapeutic gene it bears. The treatment of some diseases requires gene transfer to a stem cell, a cell that actively divides to create many new cells. For example, white blood cells live for only a short time, and must be constantly replenished by the division of **precursor** cells called hematopoietic stem cells. Gene therapy to treat an immune disease affecting white blood cells would thus require targeting these rapidly dividing cells. Researchers use a different kind of virus to accomplish this: retroviruses, so called because they contain RNA (a different kind of genetic material) rather than DNA.

When a retrovirus infects a cell, it converts its RNA to DNA and inserts it into the chromosome of the target cell. As the cell subsequently copies its own DNA during cell division, it copies the viral DNA as well, so that all of the progeny cells contain the retroviral DNA. At some later time, the viral DNA can liberate itself from the chromosome, direct the manufacture of many new viruses, and go on to repeat its life cycle. Recombinant retroviruses are engineered so that they can enter the target cell's chromosome, but become trapped there, unable to liberate themselves and continue their life cycle. Because all progeny cells still carry the recombinant retrovirus, they will also carry the therapeutic gene.

This is a great advantage over adenoviruses as a tool for gene delivery to dividing cells, but retroviruses have some drawbacks as well. They can only infect cells that are dividing quickly, and in most cases this infection must be carried out in the laboratory. Cells must be removed from the patient, infected with the recombinant retrovirus, grown for several weeks in the lab, and then reintroduced to the patient's body. This process, called **ex vivo** gene transfer, is extremely expensive and labor intensive. Nonetheless, this form of gene therapy has been used in one of the most successful clinical applications to date, the treatment of two patients with severe combined immune deficiency (SCID) caused by a defect in the adenosine deaminase gene.

Before treatment, these patients had essentially no immune system at all, and would have been required to live as "bubble children," completely isolated in a sterile environment. While their treatment did not completely cure their genetic disorder, it restored their immune systems enough to allow them to leave their sterile isolation chambers and live essentially normal lives. Many other viruses are being engineered for application to gene delivery, including adeno-associated virus, herpes simplex virus, and even extensively modified forms of the human immunodeficiency virus (HIV), to name just a few.

Many researchers are also exploring nonviral methods for gene delivery. One of the most successful of these methods consists of coating the therapeutic DNA with specialized fat molecules called lipids. The resulting small fatty drops called vesicles can then be injected or inhaled to deliver the DNA to the target tissue. Many different lipid formulations have been tested and different formulations work better in different tissues. These approaches have the great advantage that they do not stimulate the serious immune response that some viral **vectors** do. However, in general, these nonviral methods are not as efficient as viruses at transferring DNA to the target cells. No clearly superior method for gene delivery has yet emerged, and scientists are still

actively developing both viral and nonviral methods. It is likely that many different methods will eventually be used, with each method specifically tailored to work best in a specific tissue or organ of the body.

Longevity of Gene Expression

One of the most challenging problems in gene therapy is to achieve long-lasting expression of the therapeutic gene, also called the transgene. Often the virus used to deliver the transgene causes the patient's body to produce an immune response that destroys the very cells that have been treated. This is especially true when an adenovirus is used to deliver genes. The human body raises a potent immune response to prevent or limit infections by adenovirus, completely clearing it from the body within several weeks. This immune response is frequently directed at proteins made by the adenovirus itself.

To combat this problem, researchers have deleted more and more of the virus's own genetic material. These modifications make the viruses safer and less likely to raise an immune response, but also make them more and more difficult to grow in the quantities necessary for use in the clinic. Expression of therapeutic transgenes can also be lost when the regulatory sequences that control a gene and turn it on and off (called **promoters** and enhancers) are shut down. Although inflammation has been found to play a role in this process, it is not well understood, and much additional research remains to be done.

promoters DNA sequences to which RNA polymerase bind to begin transcription

Examples of Gene Therapy Applications

There are many conditions that must be met in order to allow gene therapy to be possible. First, the details of the disease process must be understood. Of course, scientists must know exactly what gene is defective, but also when and at what level that gene would normally be expressed, how it functions, and what the regenerative possibilities are for the affected tissue. Not all diseases can be treated by gene therapy. It must be clear that replacement of the defective gene would benefit the patient. For example, a mutation that leads to a birth defect might be impossible to treat, because irreversible damage will have already occurred by the time the patient is identified. Similarly, diseases that cause death of brain cells are not well suited to gene therapy: Although gene therapy might be able to halt further progression of disease, existing damage cannot be reversed because brain cells cannot regenerate. Additionally, the cells to which DNA needs to be delivered must be accessible. Finally, great caution is warranted as gene therapy is pursued, as the body's response to high doses of viral vectors can be unpredictable. On September 12, 1999, Jesse Gelsinger, an eighteen-year-old participant in a clinical trial in Philadelphia, became unexpectedly ill and died from side effects of liver administration of adenovirus. This tragedy illustrates the importance of careful attention to safety regulations and extensive experiments in animal model systems before moving to human clinical trials.

Muscular Dystrophies. Duchenne and other recessive muscular dystrophies are well suited in many ways for gene therapy. These are loss-of-function recessive genetic diseases caused by mutations in the dystrophin gene or in genes for other structural muscle proteins. The normal levels of these pro-



plasmid a small ring of DNA found in many bacteria

immunogenicity likelihood of triggering an immune system defense

teins are known, as are many of the ways that they function in the muscle cell. There is ample evidence in animal model systems that these diseases can be cured by delivery of functional copies of the gene. This is true in large part because muscle tissue has a tremendous capacity for repair and regeneration, so one could imagine that the heavily damaged muscle could repair itself after successful gene transfer. Muscle tissue is also an excellent target for gene transfer.

Several different approaches have been used to transfer DNA to muscle. The most straightforward approach is the direct intramuscular injection of DNA in a circular form called a **plasmid**. The advantage to this approach is that it induces little to no immune response, although the overall number of cells expressing the gene is fairly low. In contrast, recombinant adenoviruses are extremely efficient at transferring genes to muscle, but give rise to a potent immune response that results in only short-term expression of the transferred genes. Because the efficiency of adenoviral transfer is so great, huge efforts are underway to reduce the **immunogenicity** of these vectors. These efforts have produced some significantly improved vectors, and research is now focusing on developing methods to prepare the large quantities necessary for clinical use. Adeno-associated virus combines the extremely high efficiency of adenoviral transfer with the very low immunogenicity of direct DNA transfer. However, this virus has a rather small capacity to carry DNA, so small that it cannot carry the dystrophin gene (one of the largest genes known), which is needed to treat Duchenne muscular dystrophy.

From these examples, it should be clear that many different approaches to gene therapy for muscular dystrophy have been tried, but that each approach suffers from one or more key shortcomings. In addition, all of these approaches to treat muscular dystrophy face one common problem: Although it is easy to transfer genes to a small part of a single muscle, simultaneously delivering a gene to all parts of all the muscles of the body is impossible with today's technology.

Hemophilia and Sickle Cell Disease. Because of the difficulty in treating diseases such as muscular dystrophy, many researchers have chosen to focus on genetic diseases that may be easier to treat, particularly those resulting from the lack of proteins freely dissolved in the bloodstream. Hemophilia is one such disorder, caused by a lack of blood-clotting proteins. Such patients have long been treated by the infusion of the missing clotting proteins, but this treatment is extremely expensive and requires almost daily injections. Gene therapy holds great promise for these patients, because replacement of the gene that makes the missing protein could permanently eliminate the need for protein injections. It really does not matter what tissue produces these clotting factors as long as the protein is delivered to the bloodstream, so researchers have tried to deliver these genes to muscle and to the liver using several different vectors. Approaches using recombinant adenoviruses to deliver the clotting factor gene to the liver are especially promising, and tests have shown significant clinical improvement in a dog model of hemophilia.

Gain-of-function genetic diseases present a very different sort of challenge because the mutant gene or genes create a new biological activity that actively interferes with the normal functioning of the cell. An example of

such a disorder is sickle cell disease. Patients suffering from this disease have a defective hemoglobin protein in their red blood cells. This defective protein can cause their red blood cells to be misshapen, clogging their blood vessels and causing extremely painful and dangerous blood clots. Most of our genes make an RNA transcript, which is then used as a blueprint to make protein. In sickle cell disease, the transcript of the mutant gene needs to be destroyed or repaired in order to prevent the synthesis of mutant hemoglobin.

The molecular repair of these transcripts is possible using special RNA molecules called **ribozymes**. There are several different kinds of ribozymes: some that destroy their targets, and others that modify and repair their target transcripts. The repair approach was tested in the laboratory on cells containing the sickle cell mutation, and was quite successful, repairing a significant fraction of the mutant transcripts. While patients cannot yet be treated using this technique, the approach illustrates how biologically damaging molecules can be inactivated. Similar approaches are being developed to treat HIV-AIDS infections, and these may one day be used along with other antiviral therapies to treat this dreaded disease.

Cancer. Very different strategies of gene therapy are used to treat cancer. When treating diseases such as muscular dystrophy, researchers try to deliver genes without detection by the patient's immune system. When treating cancer, the object is often precisely the opposite: to stimulate a patient's immune reaction to the tumor tissue and improve its ability to fight the disease. For this reason, tumor tissue is often transformed by the new gene to produce specific activators of the immune system, such as interleukins or GM-CSF (granulocyte monocyte colony stimulating factor).

Usually, cancer cells are not recognized by the immune system because they are in many ways identical to the patient's normal cells. These stimulating factors activate the immune system and help it recognize and attack the tumor tissue. In another approach, called "suicide therapy," a gene such as the herpes simplex virus thymidine kinase gene (*HSV-TK*) is transferred to the tumor. This gene normally does not occur in the human body, and it is not metabolically active. After several rounds of gene therapy have built up high levels of TK activity in the tumor, a drug called ganciclovir is given to the patient. This drug is inactive in normal cells, but the TK gene converts it into a potent toxin, killing the tumor cells. Even nearby tumor cells that do not have the TK gene can be killed by a phenomenon called the "bystander effect." This approach not only kills tumor cells directly, but also activates the immune system to further attack the tumor.

Anticancer gene therapy is a powerful adjunct to other more traditional forms of cancer treatment. Its advantages are that it can be beneficial even if only a portion of the tumor cells receive the transferred gene, there is no need for long-term gene expression, and it works with the immune system, rather than trying to defeat it. Anticancer gene therapy is already in significant use in the clinic, and is likely to become even more commonplace in the near future.

In summary, gene therapy covers several related areas of research and clinical treatment, all using the genetic material DNA as a drug. Gene therapy is currently being used, along with other techniques, to treat cancer.

ribozymes RNA-based catalysts



One day, gene therapy may also be used to treat a variety of hereditary and nonhereditary diseases, ranging from loss-of-function disorders such as muscular dystrophy and hemophilia, to gain-of-function disorders such as sickle cell disease, to viral diseases such as HIV-AIDS. Active areas of research include improvements in the methods of gene delivery to the individual tissues and cells of the body and the modulation of the immune response to gene delivery. Many challenges remain to the successful maturation of gene therapy from the laboratory to the clinical setting. SEE ALSO CANCER; CYSTIC FIBROSIS; DISEASE, GENETICS OF; EMBRYONIC STEM CELLS; GENE DISCOVERY; GENE THERAPY: ETHICAL ISSUES; HEMOPHILIA; MUSCULAR DYSTROPHY; RETROVIRUS; RIBOZYME; SEVERE COMBINED IMMUNE DEFICIENCY; VIRUS.

Michael A. Hauser

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Gene Therapy: Ethical Issues

Gene therapy introduces or alters genetic material to compensate for a genetic mistake that causes disease. It is hoped that gene therapy can treat or cure diseases for which no other effective treatments are available. However, many unique technical and ethical considerations have been raised by this new form of treatment, and several levels of regulatory committees have been established to review each gene therapy clinical trial prior to its initiation in human subjects. Ethical considerations include deciding which cells should be used, how gene therapy can be safely tested and evaluated in humans, what components are necessary for informed consent, and which diseases and/or traits are eligible for gene therapy research.

Germ Line versus Somatic Cell Gene Therapy

Virtually all cells in the human body contain genes, making them potential targets for gene therapy. However, these cells can be divided into two major categories: germ line cells (which include sperm and eggs) and **somatic** cells. There are fundamental differences between these cell types, and these differences have profound ethical implications.

Gene therapy using germ line cells results in permanent changes that are passed down to subsequent generations. If done early in embryologic development, such as during preimplantation diagnosis and **in vitro** fertilization, the gene transfer could also occur in all cells of the developing embryo. The appeal of germ line gene therapy is its potential for offering a permanent therapeutic effect for all who inherit the target gene. Successful germ line therapies introduce the possibility of eliminating some diseases

somatic nonreproductive; not an egg or sperm

in vitro "in glass"; in lab apparatus, rather than within a living organism

from a particular family, and ultimately from the population, forever. However, this also raises controversy. Some people view this type of therapy as unnatural, and liken it to “playing God.” Others have concerns about the technical aspects. They worry that the genetic change propagated by germ line gene therapy may actually be deleterious and harmful, with the potential for unforeseen negative effects on future generations.

Somatic cells are nonreproductive. Somatic cell therapy is viewed as a more conservative, safer approach because it affects only the targeted cells in the patient, and is not passed on to future generations. In other words, the therapeutic effect ends with the individual who receives the therapy. However, this type of therapy presents unique problems of its own. Often the effects of somatic cell therapy are short-lived. Because the cells of most tissues ultimately die and are replaced by new cells, repeated treatments over the course of the individual’s life span are required to maintain the therapeutic effect. Transporting the gene to the target cells or tissue is also problematic. Regardless of these difficulties, however, somatic cell gene therapy is appropriate and acceptable for many disorders, including cystic fibrosis, muscular dystrophy, cancer, and certain infectious diseases. Clinicians can even perform this therapy **in utero**, potentially correcting or treating a life-threatening disorder that may significantly impair a baby’s health or development if not treated before birth.

in utero inside the uterus

Research Issues

Scientific and ethical discussions about gene therapy began many years ago, but it was not until 1990 that the first approved human gene therapy clinical trial was initiated. This clinical trial was conducted on a rare autoimmune disorder called severe combined immune deficiency. This therapy was considered successful because it greatly improved the health and well-being of the few individuals who were treated during the trial. However, the success of the therapy was tentative, because along with the gene therapy the patients also continued receiving their traditional drug therapy. This made it difficult to determine the true effectiveness of the gene therapy on its own, as distinct from the effects of the more traditional therapy.

Measuring the success of treatment is just one challenge of gene therapy. Research is fraught with practical and ethical challenges. As with clinical trials for drugs, the purpose of human gene therapy clinical trials is to determine if the therapy is safe, what dose is effective, how the therapy should be administered, and if the therapy works. Diseases are chosen for research based on the severity of the disorder (the more severe the disorder, the more likely it is that it will be a good candidate for experimentation), the feasibility of treatment, and predicted success of treatment based on animal models. This sounds reasonable. However, imagine you or your child has a serious condition for which no other treatment is available. How objective would your decision be about participating in the research?

Informed Consent

A hallmark of ethical medical research is informed consent. The informed consent process educates potential research subjects about the purpose of the gene therapy clinical trial, its risks and benefits, and what is involved in participation. The process should provide enough information for the potential

